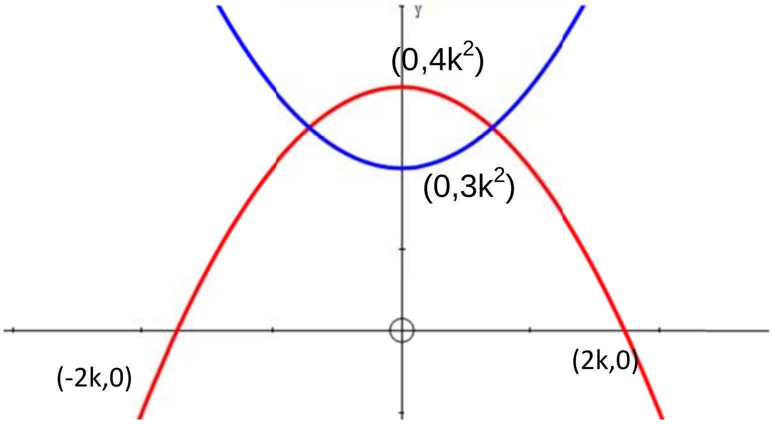
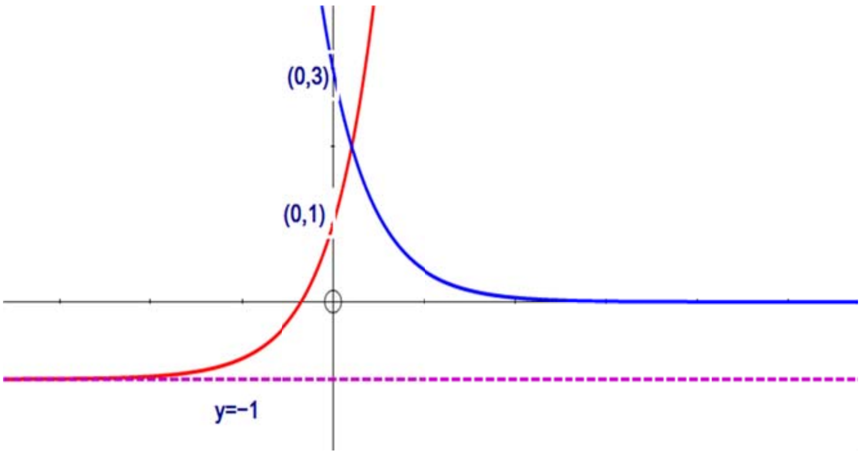
	AJC H1 Maths Prelims 2013	
1i	$\int_1^3 \frac{4}{(3x-1)}dx = \frac{4}{3}\left[\ln(3x-1)\right]_1^3$ $= \frac{4}{3}[\ln 8 - \ln 2]$ $= \frac{4}{3}\ln 4 = \frac{8}{3}\ln 2$	
	$\int \left(1-e^{-2x}\right)^2 dx = \int \left(1-2e^{-2x}+e^{-4x}\right)dx$ $= x + e^{-2x} - \frac{e^{-4x}}{4} + C$	

2ii	$(-3)^2 - 4(p-1)(p+1) < 0$ $9 - 4(p^2-1) < 0$ $-4p^2 + 13 < 0$ $4p^2 - 13 > 0$ $\left(2p+\sqrt{13}\right)\left(2p-\sqrt{13}\right) > 0$ $p < -\frac{\sqrt{13}}{2}$ or $p > \frac{\sqrt{13}}{2}$ we need $p - 1 > 0 \Rightarrow p > \frac{\sqrt{13}}{2}$	
	$y = 2(p-1)x^3 - 9x^2 + 6(p+1)x + p$ $\frac{dy}{dx} = 6(p-1)x^2 - 18x + 6(p+1)$ Increasing function $\frac{dy}{dx} > 0 \Rightarrow 6(p-1)x^2 - 18x + 6(p+1) > 0$ $(p-1)x^2 - 3x + (p+1) > 0$ for all real x From (i) $p > \frac{\sqrt{13}}{2}$	

3		
	$4k^2 - x^2 = 3x^2 + 3k^2$ $k^2 = 4x^2$ $x^2 = \frac{k^2}{4}$ $x = \frac{k}{2}$ or $-\frac{k}{2}$	
	$Area = \int_{-\frac{k}{2}}^{\frac{k}{2}} \left(4k^2 - x^2\right) - \left(3x^2 + 3k^2\right) dx$ $= 2\int_0^{\frac{k}{2}} \left(k^2 - 4x^2\right) dx$ $= 2\left[k^2 x - \frac{4x^3}{3}\right]_0^{\frac{k}{2}}$ $= 2\left[\left(\frac{k^3}{2} - \frac{4k^3}{24}\right) - 0\right]$ $= \frac{2k^3}{3}$	

4		

<table>
<tr><td></td><td>When curves intersect, $2e^{2x}-1=3e^{-2x}$
$2e^{2x}-1-\frac{3}{e^{2x}}=0$
Let $u=e^{2x}\Rightarrow 2u-1-\frac{3}{u}=0$
$2u^2-u-3=0$
$(2u-3)(u+1)=0$
$e^{2x}=\frac{3}{2}$ or $e^{2x}=-1(NA, e^{2x}>0)$
$x=\frac{1}{2}\ln\left(\frac{3}{2}\right)$</td><td></td></tr>
<tr><td>ii</td><td>$2e^{2x}-3e^{-2x}-1>0$
$2e^{2x}-1>3e^{-2x}$,
$y=2e^{2x}-1$ above $y=3e^{-2x}$
$x>\frac{1}{2}\ln\left(\frac{3}{2}\right)$</td><td></td></tr>
<tr><td></td><td>Replace x by $-\frac{t}{2}$
$-\frac{t}{2}>\frac{1}{2}\ln\left(\frac{3}{2}\right)$
$t<-\ln\left(\frac{3}{2}\right)=\ln\left(\frac{2}{3}\right)$</td><td></td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>5i</td><td>$y=x^2-\ln(2x-1)$
$\frac{dy}{dx}=2x-\frac{2}{2x-1}$
turning point $\Rightarrow 2x-\frac{2}{2x-1}=0$
$2x=\frac{2}{2x-1}$
$2x(2x-1)=2$
$2x^2-x-1=0$
$(2x+1)(x-1)=0$
$x=-\frac{1}{2}$ (NA, $x>\frac{1}{2}$) or $x=1$
<table><tr><td>x</td><td>1^-</td><td>1</td><td>1^+</td></tr><tr><td>$\frac{dy}{dx}$</td><td>-</td><td>0</td><td>+</td></tr></table></td><td></td></tr>
</table>

	Minimum point OR $\dfrac{d^2y}{dx^2}=2+\dfrac{4}{(2x-1)^2}$ At x=1, $\dfrac{d^2y}{dx^2}=2+\dfrac{4}{(2x-1)^2}=2+\dfrac{4}{1}=6>0$, minimum point	
	At x=2, $y=2^2-\ln(2\times 2-1)=4-\ln 3$ $\dfrac{dy}{dx}=2(2)-\dfrac{2}{2(2)-1}=\dfrac{10}{3}$ Gradient of normal is $-\dfrac{3}{10}$ Equation of normal is $y-(4-\ln 3)=-\dfrac{3}{10}(x-2)$ At A, $0-(4-\ln 3)=-\dfrac{3}{10}(x-2)$ $\dfrac{10}{3}(4-\ln 3)=(x-2)$ $x=2+\dfrac{40}{3}-\dfrac{10}{3}\ln 3=\dfrac{46}{3}-\dfrac{10}{3}\ln 3$ OR USE $\dfrac{0-(4-\ln 3)}{x-2}=\dfrac{-3}{10}$	
6(i)	Quota sampling Some students may report earlier than 7am or late and hence will not get any chance of being selected for the survey.	
(ii)	Conduct a stratified sampling, so that the sample will still be proportionate to the ratio of males and females of the college and thus representative of the college population, and everyone in the college will have a chance of being selected, not just those who reported after 7.00 am	
7 (i) (ii)	Let X be the height of a randomly selected plant. Since n=60>50 is large, By CLT $\overline{X} \sim N\left(100, \dfrac{5^2}{60}\right)$ approx $P(\overline{X}<99.1)=0.0816$ $P(\overline{X}>k)=0.05$ $P(\overline{X}<k)=0.95$ Min k =102.16 = 102 (3sf)	

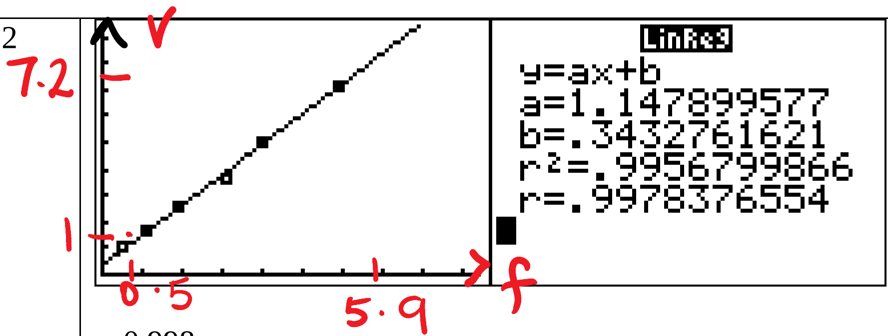
8(i)	$P(\text{win}) = P(WW) + P(WLW) + P(LWW)$ $= 0.6 \times 0.7 + 0.6 \times 0.3 \times 0.6 + 0.4 \times 0.5 \times 0.6$ $= 0.648$	
(ii)	$P(\text{win in 3rd game} \mid \text{win the match})$ $= \dfrac{P(\text{win in 3rd game AND win the match})}{P(\text{win})}$ $= \dfrac{P(\text{win in 3rd game})}{P(\text{win})} = \dfrac{P(WLW) + P(LWW)}{P(\text{Win})}$ $= \dfrac{0.6 \times 0.3 \times 0.6 + 0.4 \times 0.5 \times 0.6}{0.648} = 0.35185 \approx 0.352$	
(iii)	$P(\text{only one match won in 3rd game} \mid \text{win 3 matches})$ $= \dfrac{P(\text{only one match won in 3rd game AND win 3 matches})}{P(\text{win 3 matches})}$ $= \dfrac{P(\text{1 match won in 3rd game AND 2 matches won in 2nd game})}{P(\text{win all 3 matches})}$ $= \dfrac{P(\text{Win in 3rd game, win in 2nd game, win in 2nd game}) \times 3}{P(\text{win all 3 matches})}$ $= \dfrac{\left(0.6 \times 0.3 \times 0.6 + 0.4 \times 0.5 \times 0.6\right)\left(0.6 \times 0.7\right)^2 \times 3}{\left(0.648\right)^3} = 0.443$ **OR** $3P(\text{match won in 3rd game, won in 2nd game, won in 2nd game})$ $3 \times 0.35185 \times \left(1 - 0.35185\right)^2$ **OR** Let Y be no of matches won in 3rd game, out of the 3 matches won $Y \sim B(3, 0.35185)$ P(Y=1)=0.443	

9(i)	$\bar{x}=\dfrac{185}{50}=3.7$ $\therefore$ $s^2=\dfrac{1}{50-1}\sum\left(x-\bar{x}\right)^2=\dfrac{39.2}{49}=0.8$	
(ii)	Let X denotes the napping time of a randomly chosen child, and μ be the population mean To test $H_0:\mu=4$ $H_1:\mu\neq 4$ at 2% level of significance under H_0, & since n=50 is large, by CLT, $\bar{X}\sim N\left(4,\dfrac{0.8}{50}\right)$, $Z=\dfrac{\bar{X}-4}{\sqrt{\dfrac{0.8}{50}}}\sim N\left(0,1\right)$ Reject H_0 if p-value < 0.02 From GC, p-value $=0.017706<0.02$ $\therefore$ Reject H_0 and conclude that there is **sufficient** evidence, at 2% level of significance, to suggest that the teacher's claim is not true	
(iii)	Not the same conclusion Note $H_1:\mu<4$ (teacher's new claim) and p=0.00885<0.02. Reject Ho, i.e sufficient evidence to support teacher's new claim	
(iv)	To test $H_0:\mu=\mu_0$ $H_1:\mu>\mu_0$ at 5% level of significance Under H_0, & by CLT $\bar{X}\sim N\left(\mu_0,\dfrac{0.8}{50}\right)$ $Z=\dfrac{\bar{X}-\mu_0}{\sqrt{\dfrac{0.8}{50}}}\sim N\left(0,1\right)$ Since he did not understate μ_0, H_0 is not rejected $z_{test}\leq 1.64485\Rightarrow\dfrac{3.7-\mu_0}{\sqrt{\dfrac{0.8}{50}}}\leq 1.64485$ $\mu_0\geq 3.49$	

10(i)	Let X be the number of defective phones out of n phones $X\sim B(n,0.01)$ $P(X\geq 1)>0.95$ $1-P(X=0)>0.95$ $1-\left(0.99\right)^n>0.95$ $\left(0.99\right)^n<0.05$ $n\ln 0.99<\ln 0.05$ $n>\dfrac{\ln 0.05}{\ln 0.99}=298.07$	

	So, the least value n is 299	
(ii)	Let W be the number of defective phones out of 24 phones $W \sim B(24, 0.01)$ P(carton rejected) = $P(W \geq 2)$ $=1-P(W \leq 1)$ $= 0.0238544$ $= 0.0239$	
(iii)	Let A be the number of rejected cartons out of 250 cartons $A_B \sim B(250, 0.0238544)$ Since n is large, $np = 5.9636 > 5$ $nq = 244.036 > 5$ $\therefore A_N \sim N(5.9636, 5.82134)$ approximately $P(5 < A_B < 10)$ $\approx P(5.5 < A_N < 9.5)$ $= 0.504823$ $= 0.505$ (3sf)	
11(i)	Let M be the weight of a randomly chosen male and F be the weight of a randomly chosen female tourist. $M \sim N\left(75, 10^2\right)$ and $F \sim N\left(\mu, \sigma^2\right)$ Now, $P(F < 43) = 0.1$ $P\left(Z < \frac{43-\mu}{\sigma}\right) = 0.1 \Rightarrow \frac{43-\mu}{\sigma} = -1.28155$ $\mu - 1.28155\sigma = 43$ ---(1) Also, $P(F > 55) = 0.18$ $P(F < 55) = 0.82 \Rightarrow P\left(Z < \frac{55-\mu}{\sigma}\right) = 0.82$ $\frac{55-\mu}{\sigma} = 0.915365$ $\mu + 0.915365\sigma = 55$ ---(2) Using GC, $\mu = 50$ and $\sigma = 5.46$	
(i)	Let $A = \frac{M + F_1 + ... + F_5}{6}$ $E(A) = \frac{1}{6}\left(75 + 5 \times 50\right) = 54.1666$ $Var(A) = \frac{1}{6^2}\left(100 + 5 \times 5.46^2\right) = 6.91827$ $A \sim N\left(54.1666, 6.91827\right)$ $P(A > 60) = 0.0133$	
()	Let $D = \left(F_1 + ... + F_5\right) - 3M$ $E(D) = 5 \times 50 - 3 \times 75 = 25$ $Var(D) = 5 \times 5.46^2 + 3^2 \times 10^2 = 1049.058$ $\therefore D \sim N\left(25, 1049.058\right)$ $P(D > 5 \text{ or } D < -5) = 1 - P(-5 < D < 5) = 0.909$	

(iii)	$M \sim N\left(75,10^2\right)$ and $F \sim N\left(50,5.46^2\right)$ Let C be the number of female tourist, out of 5 female tourists, that weighs more than 55kg $C \sim B\left(5,0.18\right)$ $P\left(\text{ half of them heavier than 55kg}\right)$ $=P(C=2)P(M>55)+P(C=3)P(M<55)$ $=(0.17864)(0.97724)+(0.39214368)(0.02275)$ $=0.183495 \approx 0.183$	
	$P\left(\text{female}\mid\text{heavier than 55kg}\right)=\dfrac{P\left(\text{female AND heavier than 55kg}\right)}{P\left(\text{heavier than 55}kg\right)}$ $=\dfrac{\frac{1}{3}P(F>55)}{\frac{1}{3}P(F>55)+\frac{2}{3}P(M>55)}=\dfrac{\frac{1}{3}(0.18)}{\frac{1}{3}(0.18)+(\frac{2}{3})(0.9772499)}$ $=0.0843$	

12	 r=0.998 There is a positive correlation between the *f*, the weekly sales of face masks, and *v*, the weekly sales of the bottles of Vitamin C sold. As the weekly sale of face masks increase, the weekly sales of Vitamin C bottles tends to increase.	
	$v=1.15f+0.343$ $k=1.15$ means for every **additional** 1000 face masks sold in a week, an average of 115 **additional** bottles of Vitamin C are sold in a week.	
	Use v on f: $v=1.14790\left(2.5\right)+0.34328=3.21$ **321 bottles** of Vitamins C are sold. f=2.5 is within the data range of $0.5 \le f \le 5.9$ **and** r=0.998 is close to 1. Estimate is reliable because interpolation for 2 strongly correlated variables is reliable	
	Not reliable. To estimate *f*, he should use the **regression line of** f on v to estimate *f*.(since there is no clear controlled variable,	